VISIVAULT ANALYTICS

Prairie Auto Care: Repair Operations & Financial Intelligence

Management report - what a multi-location collision & car care business can learn from its own data

⚠ **Developed by VisiVault Analytics Ltd. — Synthetic demonstration data. Not actual Ultimate Car Care Group records.**

Developed by VisiVault Analytics Ltd. · October 2026 · Management report

Prairie Auto Care Group is a FICTIONAL multi-location Calgary collision repair and car care business. Its business model (three Calgary locations with distinct roles, published hours, service menu, OEM-certified collision work, direct billing of insurers) is inspired by a public Calgary auto care group's website (retrieved 2026-10-09); every repair order, cost, technician, insurer, customer and financial figure is synthetic. No relationship with or endorsement by any real company is implied.

Important notice

> This report analyses a **fictional** auto care group built on synthetic data to demonstrate VisiVault Analytics' capabilities. None of the operational or financial figures describe any real company. The analyses are designed to complement - not replace - a repair business's existing shop-management, insurance-estimating, parts-ordering and accounting systems.

Interactive dashboard: prairie-auto-care-bi.streamlit.app - 15 pages, the same synthetic data as this report.

1. Executive summary

NET REVENUE (TTM)	EBITDA (TTM)	COLLISION KEYS-TO-KEYS	PARTS WAITING
$14.26M	**$1.25M**	**11.6 d**	**9.5 d**
3 branches, 10,447 ROs	8.7% of revenue	25 days from first call	largest collision delay

ORIGINAL PROMISE MET	TECHNICIAN PRODUCTIVITY	INSURER RECEIVABLES	BASE FORECAST OP. PROFIT
62%	**63%**	**$573,561**	**$1.46M**
80% after revision	util. 60% x eff. 106%	27 days to pay	Oct-26 to Sep-27 (FORECAST)

- Parts waiting is the largest delay in collision repair: 9.5 days on average between approval and parts complete, vs 1.8 days waiting for a technician.
- Supplements (hidden damage) occur on 25% of collision jobs; only 62% of collision jobs meet the ORIGINAL promise date, 80% meet the revised promise.
- Late jobs score 3.73 / 5 in customer surveys vs 4.51 for on-time jobs.
- 1736 ROs (16.6%) did not cover their direct costs plus shop overhead ($88 per clocked hour) in the last 12 months - $355,820 short; warranty comebacks cost $22,291.
- Technician utilisation 60%, efficiency 106%, productivity 63%; overtime 4.3% of available hours.
- Non-DRP insurers take 4.0 days to approve vs 0.7 for DRP insurers (fictional insurers).

2. Public information vs illustrative findings

Only the business-model facts below come from public information; they shaped the structure of the fictional group (three locations, roles, hours, services). Everything else in this report is computed from synthetic data and is illustrative.

Grounded in public information (a Calgary auto care group's website)	How the fictional demo uses it
Three locations with distinct roles	Fictional branches: SE Concierge (newest; detailing, protection, light collision), Central Collision Centre (collision hub), North Full-Service (collision + mechanical, tires, glass)
SE location recently opened	SE Concierge opens 1-May-2025 in the simulation and ramps up over 9 months (assumption; actual opening date not used)
Opening hours	Technician schedules and appointment slots follow the published hours (Mon-Fri 8-5; Sat 10-1 at SE and North)
Service menu	Service types and revenue lines: collision, refinish, PDR, glass, detailing, paint protection, tires, diagnostics, maintenance, mechanical & alignment, ADAS calibration; towing & rental handled as sublet / pass-through
OEM-certified collision + insurer direct billing	Most collision work is insurer-paid with direct billing; fictional insurers in DRP / non-DRP categories

Grounded in public information (a Calgary auto care group's website)	How the fictional demo uses it
Lifetime workmanship warranty	Comebacks (warranty rework) are repaired at no charge and costed to warranty rework

Illustrative findings from SYNTHETIC data (methods, not facts about any real business)
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Non-DRP insurers take 4.0 days to approve vs 0.7 for DRP insurers (fictional insurers).

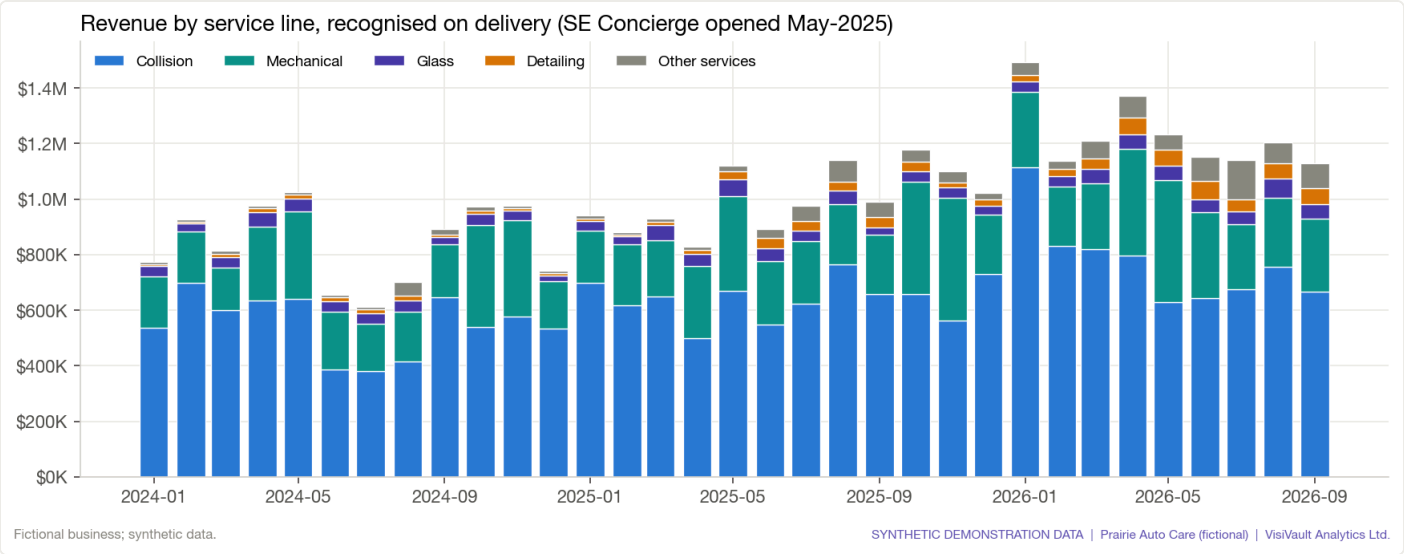
3. Business challenges this platform addresses

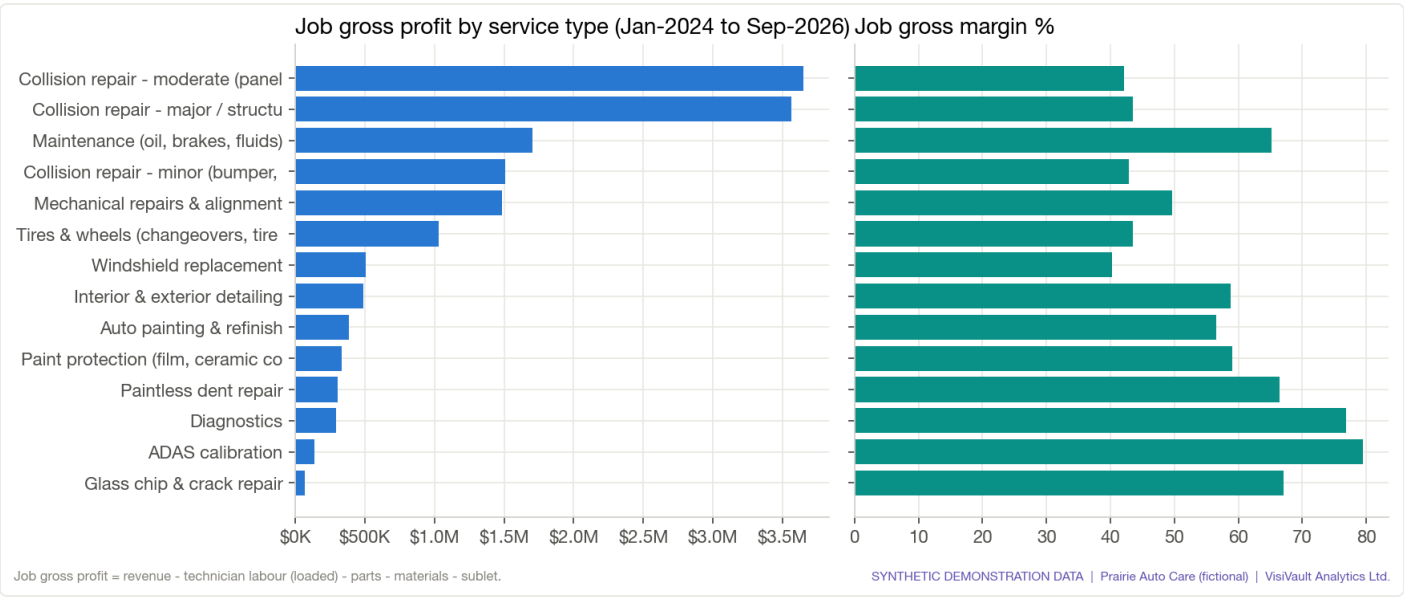
- **Cycle time and promises:** parts availability, insurer approvals and supplements push vehicles past their promised date - the strongest driver of customer satisfaction.
- **Technician capacity:** collision demand peaks with snow and winter driving, tires peak twice a year, summer is quieter; labour is the largest controllable cost.
- **Job profitability:** fixed customer-pay quotes, parts price changes, discounts and comebacks erode margin job by job - often invisible in monthly totals.
- **Insurer cash cycle:** direct billing means most collision revenue is collected weeks after delivery; documentation quality drives payment speed.
- **Multi-branch economics:** work moves between branches (e.g. a concierge site sending collision repairs to the main shop); fair branch results need transfer pricing, eliminations and transparent overhead allocation.

4. Platform scope and KPIs

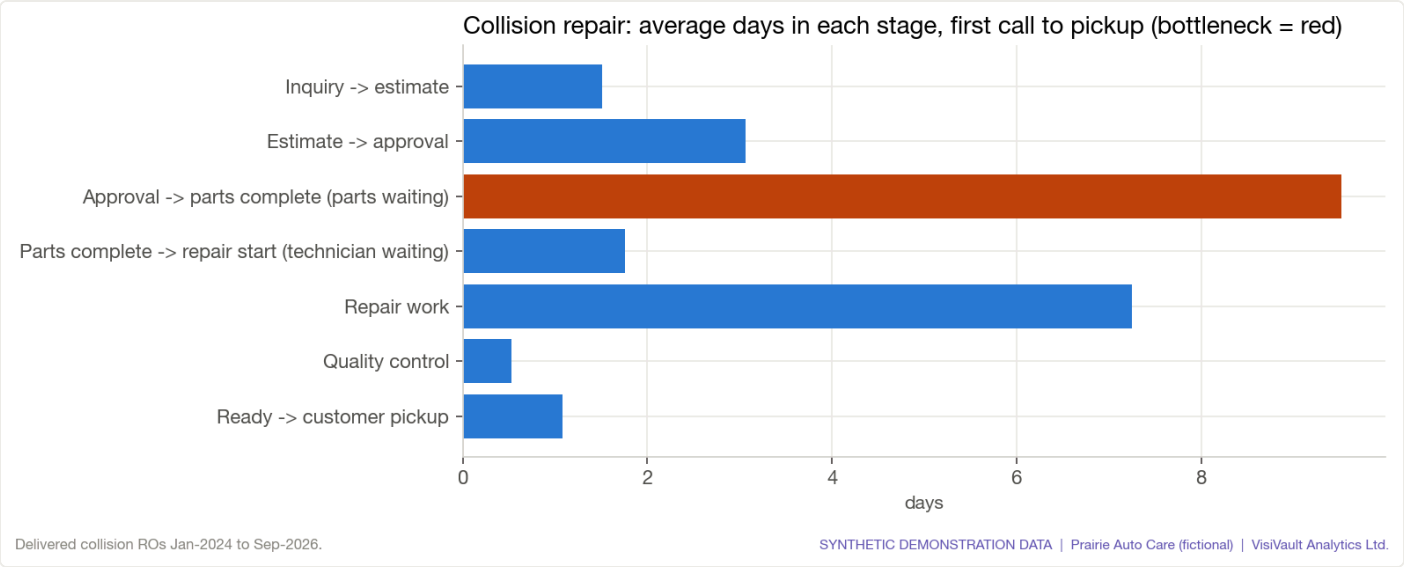
Module	Key performance indicators
Job profitability	Gross profit and fully loaded profit per RO by service, vehicle, payer and branch; estimate-to-actual cost variance; unprofitable ROs; comeback cost
Technician productivity	Utilisation, efficiency and productivity (separately defined); overtime; mix-adjusted efficiency and comeback indices by role
Repair cycle time	Keys-to-keys and first-call-to-delivery; time in each stage (approval, parts, technician, repair, QC, pickup); on time vs original and revised promise
Parts & suppliers	Fill rate, emergency orders, FIFO aging, turnover, carrying cost; supplier on-time %, lead time, backorders, returns
Insurance claims	Approval time, approved vs estimated, supplement rate, receivables ageing and payment days by insurer, claim admin workload
Branch & finance	Branch P&L with eliminations and allocated overhead, branch cash flow, three statements, ratios, 12-month forecast and scenarios

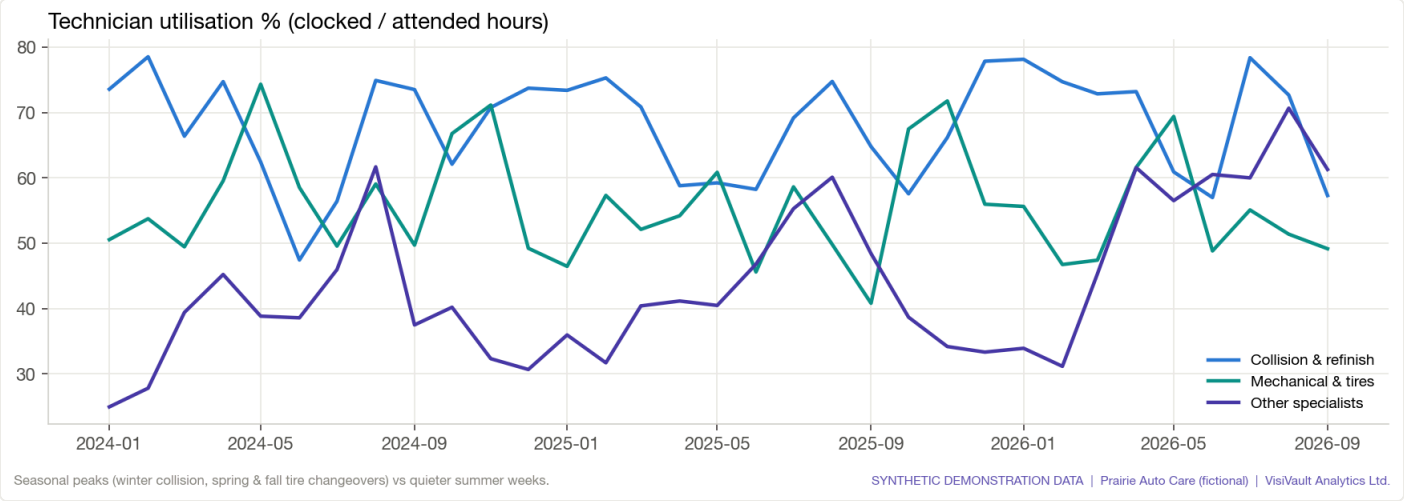
5. Analytical findings (synthetic data)





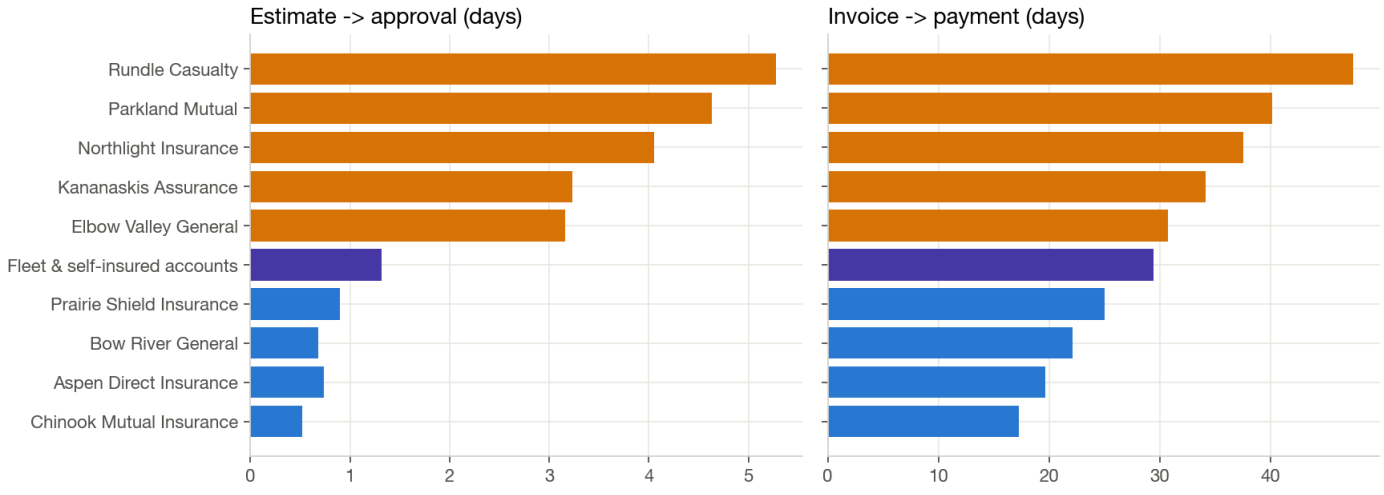
Job gross margin averages 47.4% before overhead; 1,736 ROs (16.6%) did not cover direct cost plus shop overhead absorbed at $88 per clocked hour ($355,820 short in 12 months).





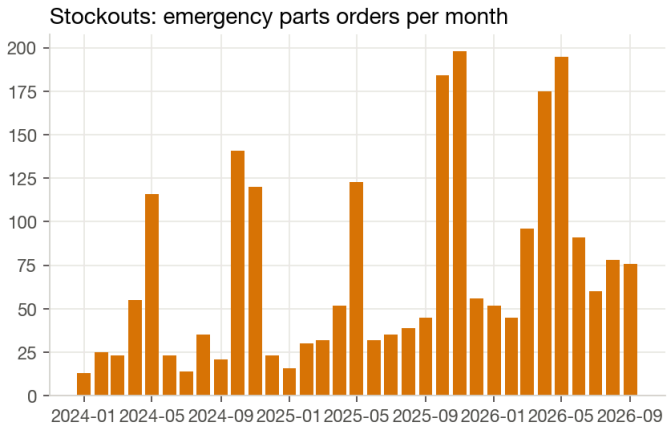
Utilisation 60%, efficiency 106%, productivity 63%; overtime 4.3% of available hours. Technicians are compared with the expected efficiency and comeback rate for their own job mix and role, not ranked on raw totals.

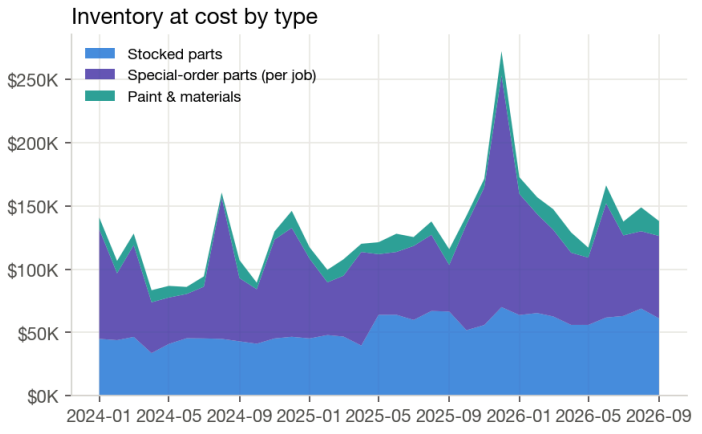
6. Insurance claims, parts and branches



FICTIONAL insurers. Blue = DRP, amber = non-DRP, purple = fleet / self-insured.

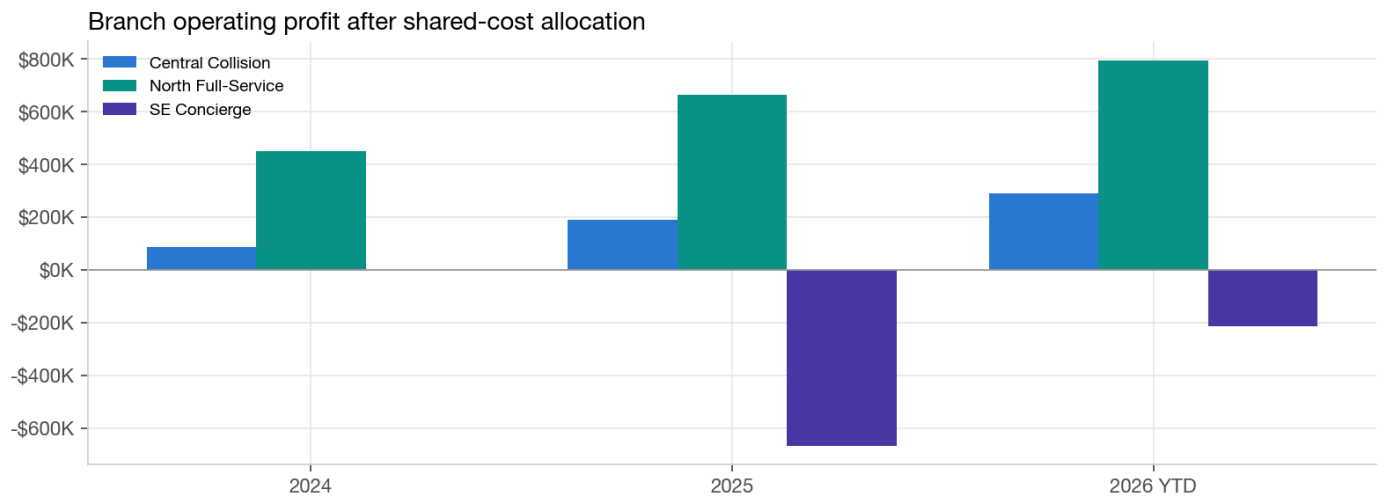
SYNTHETIC DEMONSTRATION DATA | Prairie Auto Care (fictional) | VisiVault Analytics Ltd.





Current reorder points are flat (non-seasonal) - tire seasons drive stockouts.

SYNTHETIC DEMONSTRATION DATA | Prairie Auto Care (fictional) | VisiVault Analytics Ltd.

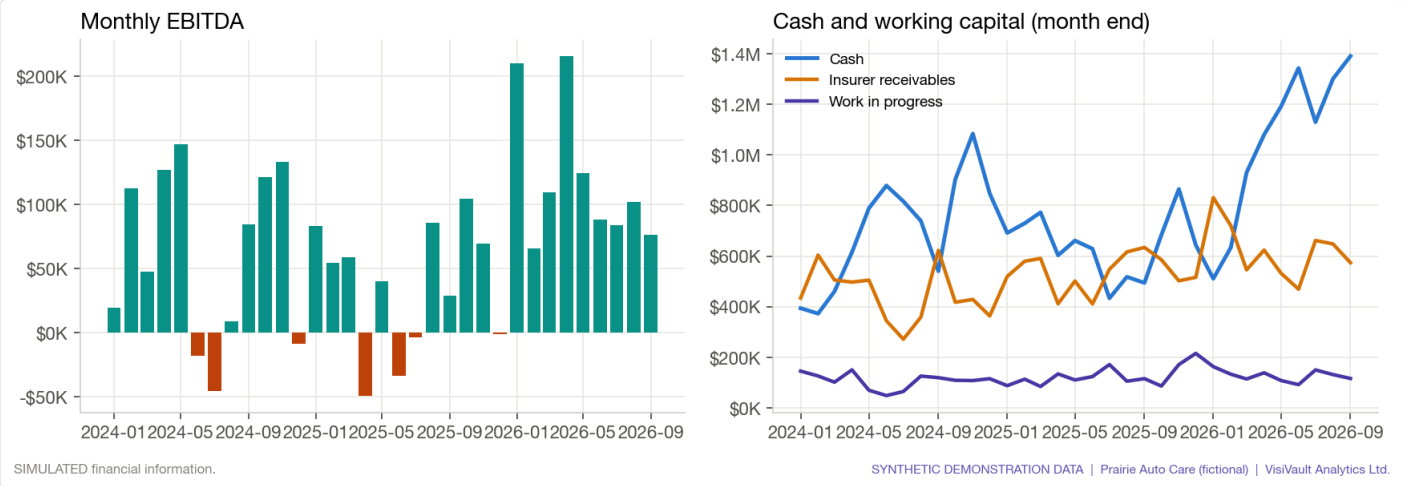


Interbranch work at transfer price; head-office costs allocated by documented drivers.

SYNTHETIC DEMONSTRATION DATA | Prairie Auto Care (fictional) | VisiVault Analytics Ltd.

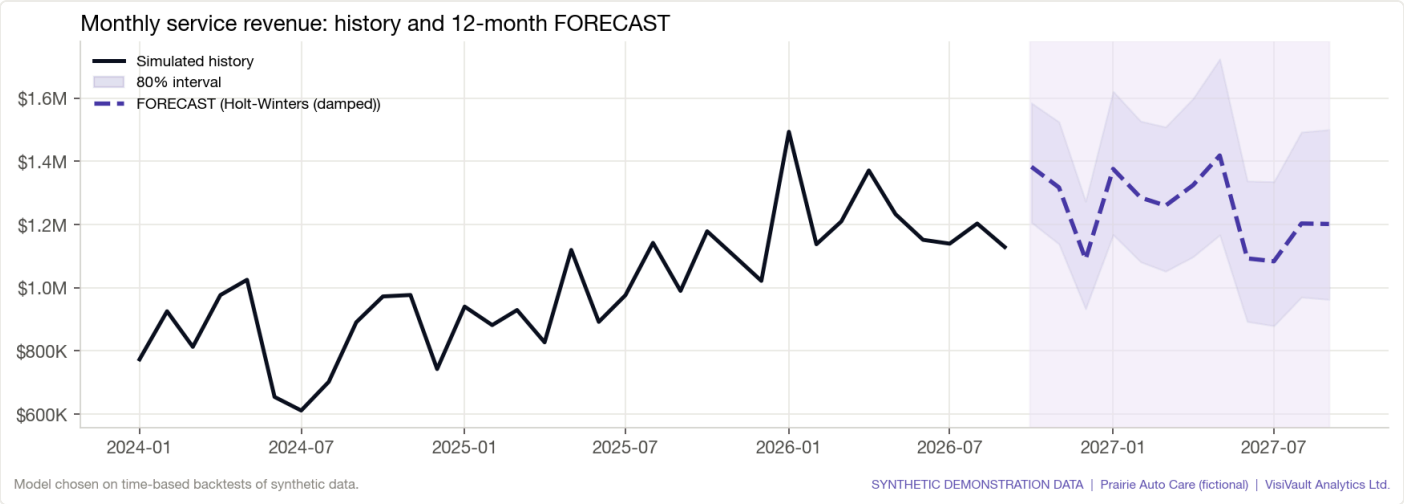
7. Financial performance (SIMULATED)

CAD	FY2024	FY2025	YTD Sep-2026
Collision repair revenue	6,589,833	7,680,612	6,935,225
Mechanical service revenue	2,741,700	3,154,373	2,601,799
Detailing revenue	146,204	288,569	421,992
Glass repair revenue	429,996	487,070	444,536
Other service revenue	156,605	386,951	662,943
Total net revenue	10,002,471	11,921,081	10,978,567
Technician labour	(2,305,014)	(2,921,898)	(2,578,143)
Parts	(3,719,165)	(4,332,510)	(3,900,169)
Paint & materials	(244,902)	(301,831)	(290,273)
Sublet & towing	(132,367)	(164,193)	(130,025)
Cost of sales	(6,410,323)	(7,728,324)	(6,909,613)
Gross profit	3,592,148	4,192,758	4,068,955
Operating expenses	(2,863,530)	(3,756,025)	(2,993,831)
EBITDA	728,618	436,733	1,075,123
Depreciation	(190,900)	(251,583)	(204,379)
Interest	(32,439)	(43,139)	(26,179)
Income taxes	(56,214)	(15,621)	(134,250)
Net income	449,065	126,389	710,315



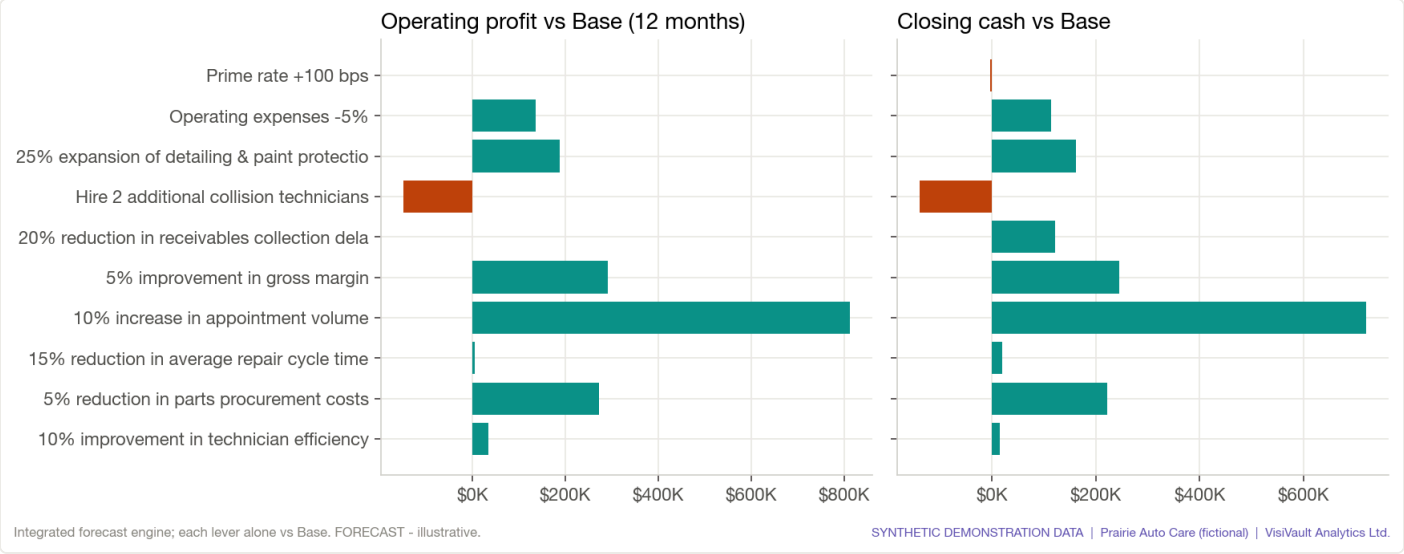
Work in progress is carried at cost until delivery; insurer receivables average 28 days. Cash conversion cycle -5 days (parts 10 + WIP 5 + receivables 17 - payables 38). All 12 financial validation checks pass, including the Excel model's own recalculation (max difference vs Python $0.000000).

8. Financial forecasting capabilities (FORECAST)



method	WAPE	bias_pct	PI80_coverage
Seasonal naive	17.7%	-17.7%	67%
Recent level x seasonal index	9.9%	+0.6%	67%
Holt-Winters (damped)	11.1%	-6.5%	50%
SARIMAX	12.6%	+2.3%	83%
XGBoost	11.5%	+9.4%	0%

Six models: revenue, RO volume / billed-hours demand, repair cycle time (best: Ridge regression, MAE 6.1 days vs 6.5 for the baseline), parts demand, technician capacity and expected cash collections. Each is compared against a simple baseline with time-based validation. **Model performance on synthetic data does not establish real-world prediction accuracy.**



Collision and mechanical revenue are capacity-constrained in the forecast: a 10% efficiency gain changes revenue by only $24,237 because demand - not technician capacity - limits sales most months; the gain shows up as lower overtime.

9. Improvement opportunities and recommendations

#	Area / action	Op. profit, 12m	Cash, 12m	Effort / timing
1	**Job profitability** Price and quote review for jobs that do not cover shop overhead; teardown before fixed customer-pay quotes	$142,328	$128,095	Low / 0-3 months
2	**Growth** Grow detailing & paint protection at SE Concierge (insurer-referral and dealer packages)	$187,364	$161,876	Medium / 3-9 months
3	**Parts procurement** Preferred-supplier agreement and monthly price audit on the top OEM / aftermarket suppliers	$163,169	$133,809	Medium / 3-6 months
4	**Technician productivity** Flat-rate incentive and training to lift efficiency - paired with demand growth	$34,017	$16,193	Medium / 3-6 months
5	**Parts inventory** Seasonal, service-level reorder points for stocked parts (tires, oil, filters, brakes)	$13,048	$13,048	Low / 0-3 months
6	**Quality** Pre-delivery QC checklist and comeback root-cause review (lifetime warranty)	$7,802	$7,802	Low / 0-6 months
7	**Repair cycle time** Parts-first scheduling with daily supplier escalation for collision jobs	$5,297	$20,408	Low / 0-3 months
8	**Insurance receivables** Same-day electronic invoicing with complete photo & supplement files; weekly follow-up of items past terms	$0	$122,626	Low / 0-3 months

1. Price and quote review for jobs that do not cover shop overhead; teardown before fixed customer-pay quotes

Estimate: operating profit $142,328, cash $128,095 (12 months). **How calculated:** 1736 ROs (16%) fell $355,820 short of direct cost + shop overhead ($88 per clocked hour) in 12 months; largest gaps: Interior & exterior detailing ($282,249), Paintless dent repair ($21,755), Paint protection (film, ceramic coating) ($10,680). Assume 40% recovered through minimum charges, package pricing and quote discipline. **Assumptions:** Customers accept minimum labour charges / revised quotes; goodwill discounts approved by the branch manager only. **KPIs:** Fully loaded profit per RO, unprofitable RO count, estimate-to-actual labour variance

2. Grow detailing & paint protection at SE Concierge (insurer-referral and dealer packages)

Estimate: operating profit $187,364, cash $161,876 (12 months). **How calculated:** Engine re-run with detailing & other services +25% (with installer labour). **Assumptions:** Demand exists at current prices; installer capacity added in line with volume. **KPIs:** SE revenue, SE contribution, PPF backlog days

3. Preferred-supplier agreement and monthly price audit on the top OEM / aftermarket suppliers

Estimate: operating profit $163,169, cash $133,809 (12 months). **How calculated:** Special-order parts are ~82% of parts spend. Engine re-run with parts cost -3%. **Assumptions:** Volume rebate or discount of 3% on special-order parts; no change in quality or lead time. **KPIs:** Parts margin %, price variance vs estimate

4. Flat-rate incentive and training to lift efficiency - paired with demand growth

Estimate: operating profit $34,017, cash $16,193 (12 months). **How calculated:** Engine re-run with efficiency +10%: only $24,237 extra revenue because collision demand rarely exceeds capacity; the benefit is mainly lower overtime. Efficiency

pays off when combined with volume growth. **Assumptions:** Demand-constrained most months; capacity binds only in winter peaks. **KPIs:** Efficiency, overtime %, productivity

5. Seasonal, service-level reorder points for stocked parts (tires, oil, filters, brakes)

Estimate: operating profit $13,048, cash $13,048 (12 months). **How calculated:** 1,306 emergency (stockout) orders in 12 months, $202,961 bought at a ~12% premium ($21,746). 121 branch-SKUs need a higher seasonal reorder point. Assume 60% of the premium avoided. **Assumptions:** Reorder points from reorder_point_recommendations.csv (95/90/85% service by ABC class); weekly review kept. **KPIs:** Fill rate, emergency orders, inventory turnover, aged stock

6. Pre-delivery QC checklist and comeback root-cause review (lifetime warranty)

Estimate: operating profit $7,802, cash $7,802 (12 months). **How calculated:** Warranty comebacks cost $22,291 in parts & labour over 12 months (1.5% of ROs). Assume 35% fewer comebacks. **Assumptions:** Comeback rates compared with the expected rate for each technician's job mix (no raw ranking). **KPIs:** Comeback rate, rework cost, CSAT

7. Parts-first scheduling with daily supplier escalation for collision jobs

Estimate: operating profit $5,297, cash $20,408 (12 months). **How calculated:** Parts waiting averages 9.5 days per collision job vs 1.8 days waiting for a technician. Weakest on-time suppliers: Import Specialty Parts (fictional) (43%), Westline Glass Distribution (fictional) (46%). Engine re-run with cycle time -15% (less work in progress, fewer waiting customers lost). **Assumptions:** Vehicles dropped off only when parts are confirmed; backorders escalated within 24 h; alternative certified parts approved faster. **KPIs:** Keys-to-keys days, parts waiting days, on-time vs original promise, CSAT

8. Same-day electronic invoicing with complete photo & supplement files; weekly follow-up of items past terms

Estimate: operating profit $0, cash $122,626 (12 months). **How calculated:** Insurer receivables $573,561 at 30-Sep-2026; average payment 27 days (non-DRP insurers slower: 38 days). Engine re-run with collection delays -20%. **Assumptions:** Most delay is documentation and follow-up, not insurer policy; fictional insurers only. **KPIs:** Insurer DSO, receivables > 60 days, short-pays

10. Implementation requirements

- Read-only exports (CSV / API / scheduled reports) from the shop-management system: repair orders with status history, technician time clock, parts lines.
- Estimating and insurer data already exchanged with insurers: estimates, supplements, approvals; invoice and payment remittances.
- Accounting: monthly trial balance / GL, AR and AP ageing; payroll summary by branch; fixed-asset register.
- A named data contact (2-3 h/week) and a monthly 60-minute management review.

11. Risk considerations

- **Data quality:** clock-on discipline and RO status timestamps vary; the diagnostic measures completeness before any KPI is published.
- **Fairness:** technician metrics are mix- and role-adjusted and shown with evidence levels; they support coaching, not discipline.
- **Privacy & security:** customer personal information is excluded or pseudonymised; data stays in Canada (PIPEDA / Alberta PIPA); read-only access.
- **Model risk:** forecasts are re-validated on real history before use; scenario results are decision aids, not guarantees.
- **Insurer relationships:** claim analytics are for internal process improvement; no insurer is characterised publicly.

12. Proposed 90-day pilot

When	Phase	Deliverables
Weeks 1-2	Diagnostic	Data-access review; export mapping from the shop-management, estimating and accounting systems; data-quality report; KPI definitions agreed
Weeks 3-6	Build	Job profitability and technician productivity views on real data; cycle-time stage timestamps; branch P&L with agreed transfer pricing and allocations
Weeks 7-10	Validate	Reconcile to the general ledger and payroll; review with branch managers and estimators; refine fairness adjustments for technician metrics
Weeks 11-13	Embed	Monthly management pack, weekly cycle-time / parts-delay list, 12-month forecast with scenarios; handover and next-phase plan

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